

Fault Detection for Medical Body Sensor Networks Under Bayesian Network Model

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Abstract—We propose a Bayesian network based method for the fault diagnosis problem of medical body sensor networks used to collect physiological signs to monitor the health of patients. We formalize a Bayesian network to describe the body sensor network considering both the spatial and temporal correlation in measurements at different sensors. Then we give the theoretical analysis of the fault detection, false alarm of this method, and the error probability after executing the fault diagnosis algorithm. Finally, Experiments carried out on synthetic medical datasets by injecting faults into real medical datasets show that the simulation performance matches the theoretical analysis closely, and the proposed approach possesses a good detection accuracy with a low false alarm rate.

Index Terms—fault diagnosis; body sensor networks; Bayesian network

I. INTRODUCTION

The medical body sensor network (BSN) uses numerous small wireless devices attached to or implanted in the body of a person to collect various vital signs such as heart rate (HR), oxygen saturation (SpO₂), blood pressure (BP), etc, and transmit collected data to a central device for processing [7]. Due to restriction of resources, sensor data are unreliable, which leads to faulty diagnosis results, false alarms, and an unreliable monitoring. So fault detection is extremely important to ensure the reliability and accuracy of sensor data before the medical diagnosis-making process [12].

Fault detection for wireless sensor networks have been studied for many years [1]-[8]. But we can use these methods directly for BSNs. To the best of our knowledge, the existing works related to fault detection for BSNs are [7]-[12]. Kim *et al.* in [9]-[10] used rule-based method for simple fault detection in BSNs. However, these approaches are not accurate, because it is difficult to find out a right threshold value of correlations.

Salem *et al.* in [12] used a Kernel function to estimate the distribution of the distance between a sensor reading to the mean of training data. A data instance is diagnosed as faulty if the probability of the calculated distance is very low. Kim *et al.* in [9] and [10] also gave another approach for estimating a Gaussian distribution to compute fault probability for new node reading. However those statistical based approaches may have bad performance when the faulty readings are normal values without a low probability. And it is a challenge that the training data may contain faults, which disrupts the estimated values for these statistical parameters.

Salem *et al.* in [7] tried to use a linear regression to estimate the reading of a sensor, e.g. HR, by values of its neighbour sensors. A reading is considered as a fault if its deviation to the estimated value larger than a threshold. However, human vital signs may not have linear correlations, and it can not decide which reading is faulty when some fault occurs.

Bayesian network (BN) based approaches can aggregate information from different variables to estimate the sensor reading and give a decision whether such reading is faulty. The main advantage of Bayesian network based techniques is that it can capture both spatial and temporal correlations among sensor nodes [14]. Janakiram *et al.* in [13] developed the framework of outlier detection for wireless sensor networks with the Bayesian network model. However, the model they used only contained sensor readings without the ground truth value, it can not yet deal with the decision threshold scheme.

In this paper, we formalize a Bayesian network model to represent conditional dependencies among both the hidden ground truth values of physiological signs and the measured sensor readings. A sensor measurement is considered as faulty if the probability of the ground truth value equalling to such reading under all sensor measurements of the BSN is less than a preselected threshold. The experiment results show that the number of errors can be reduced by 60% using this Bayesian network fault detection method.

The main contributions of this paper are summarized as follows.

(1) We formalize a BN model to describe health monitoring with BSNs. This model contains the current, previous, succeeding and neighbour ground truth values as well as sensor readings of the diagnosed physiological sign. A sensor reading is considered as faulty if the probability of the ground truth value equalling to this reading under the previous, succeeding and neighbour sensor readings is greater than a preselected threshold.

(2) We give the theoretical performance analysis of the fault detection, false alarm, and the error probability after fault diagnosing. Besides, we give an algorithm for selecting approximately the optimal threshold which minimizing the theoretical error probability after executing the fault diagnosis method.

(3) We study the performance of the proposed approach on artificial datasets which is produced by injecting faults of varying intensity into datasets from a medical database.

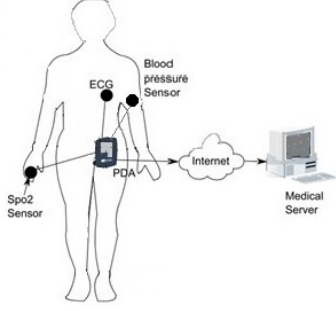


Fig. 1. The Architecture of BSNs.

The rest of this paper is organized as follows. Section II introduces system models and definitions. In section III, the fault detection method based on Bayesian network is proposed. Section IV does theoretical analysis of the proposed fault detection method. In Section V, experiments are carried out to demonstrate the performance of the proposed method. Conclusions are drawn in Section VI.

II. SYSTEM MODELS AND DEFINITIONS

Fig. 1 shows the network architecture of our considering medical deployment scenario. We use three sensors to monitor heart activities, blood pressure, respiration rate and saturation of oxygen in the arterial blood. These sensors monitor physiological signs and transmit the collected data periodically at every discrete time instance to neighboring personal server devices, such as a smartphone. Then by wireless and wired connection, these data are streamed remotely to a medical doctor's site for real time diagnosis, to a medical database for record keeping, or to the corresponding equipment that issues an emergency alert.

Bayesian network: A Bayesian network is a directed acyclic graph (DAG). Each node represents a random variable ranging over a discrete domain values. An edge describes a probabilistic dependency between a parent and a child. Besides, each node has a conditional probability table (CPT) representing the probability of such node's values under the combination values of its parents. If a node has no parent, the CPT becomes to the distribution probabilities of the node's possible values [16].

Notations: We use $Q^H = \{q_1^H, \dots, q_n^H\}$, $Q^B = \{q_1^B, \dots, q_m^B\}$ and $Q^S = \{q_1^S, \dots, q_k^S\}$ to denote respectively the discrete ground truth values of HR, BP and SpO2, and $V^H = \{v_1^H, \dots, v_n^H\}$, $V^B = \{v_1^B, \dots, v_m^B\}$ and $V^S = \{v_1^S, \dots, v_k^S\}$ the sensor measurement values of HR, BP and SpO2. Besides, we use s_t to denote the ground truth value of HR at time t .

In the following, we formalize a Bayesian network based method to detect faulty measurements for the diagnosed sensor in BSNs.

III. FAULT DETECTION

A. Fault Detection Model for BSN

We construct a Bayesian network shown in Fig.2 to model the attributes at the sensor node in BSNs. In this model, nodes H , L and N respectively represent the ground truth value of the current, last time and next time of HR. B and S are respectively ground truth values of current BPmean and SpO2. H' , L' and N' are respectively the current, last time and next time sensor readings of HR, B' and S' are respectively current sensor readings of BPmean and SpO2. Then $H' = H$ (or $B' = B$, $S' = S$) represents the scenario that the sensor correctly collects a physiological sign value, and $H' \neq H$ that the sensor faultily reports a physiological sign value.

We use b_{ij}^H to denote the conditional probability on node H' which is the observation symbol probabilities for HR. Then

$$\begin{aligned} b_{ij}^H &= P(H' = v_j^H | H = q_i^H) = P(L' = v_j^H | L = q_i^H) \\ &= P(N' = v_j^H | N = q_i^H) \quad 1 \leq i \leq n, 1 \leq j \leq n' \end{aligned} \quad (1)$$

Similarly, we use b_{ij}^B and b_{ij}^S to denote conditional probabilities on node B' and S' , that are

$$b_{ij}^B = P(B' = v_j^B | B = q_i^B) \quad 1 \leq i \leq m, 1 \leq j \leq m' \quad (2)$$

$$b_{ij}^S = P(S' = v_j^S | S = q_i^S) \quad 1 \leq i \leq k, 1 \leq j \leq k' \quad (3)$$

We use c_{ijl} to denote the conditional probability on node H , which is the causal relation from B , S to H relates to the spatial correlation between BPmean and SpO2 to HR. That is

$$c_{ijl} = P(H = q_i^H | B = q_j^B, S = q_l^S), \quad 1 \leq i \leq n, \quad 1 \leq j \leq m, 1 \leq l \leq k \quad (4)$$

The causal relations from H to L and N relate to the temporal correlation of HR data. We use a_{ij} to denote the transition probability with $a_{ij} = P(s_{t+1} = q_j^H | s_t = q_i^H)$, $1 \leq i, j \leq n$. Then the conditional probability on node N is

$$a_{ij} = P(S = q_j | X = q_i) \quad (5)$$

the conditional probability on node L is

$$\bar{a}_{ij} = P(L = q_j^H | H = q_i^H) = \begin{cases} \frac{a_{ji}}{\sum_{r=1}^n a_{ri}}, & \text{if } \sum_{r=1}^n a_{ri} > 0 \\ \frac{1}{n}, & \text{otherwise} \end{cases} \quad (6)$$

In addition, we use π_i^H , π_i^B and π_i^S to denote respectively the distribution probability of HR, BPmean and SpO2 with $\pi_i^H = P(H = q_i^H)$, $\pi_i^B = P(B = q_i^B)$ and $\pi_i^S = P(S = q_i^S)$.

B. Fault Detection Method

Let X denote the variables set $\{L', N', S', B'\}$, we assume that there is an order of the n^2mk combination values $\{\nu_i | 1 \leq i \leq n^2mk\}$ of variables in X , we use $X = \nu_h$ to denote that variables in X are assigned the h th combination values $\{v_j^H, v_u^H, v_d^S, v_f^B\}$, which is

$$L' = v_j^H, N' = v_u^H, S' = v_d^S, B' = v_f^B \quad (7)$$

Since the ground truth values of physiological signs are unobservable, so given the current sensor reading $H' = v_i^H$,

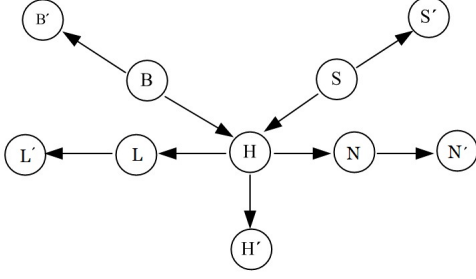


Fig. 2. A Bayesian Network for BSNs.

the task of fault detection is to determine a value $\mathcal{H}$ for H considering both the spatial correlation—the readings S' , B' of neighbouring sensors, and the temporal correlation—the last and next readings L' and N' of the same sensor. Theorem 1 tells us the result.

Theorem 1. Given $H' = v_i^H$ and variables in X assigned the h th combination ν_h with $L' = v_j^H$, $N' = v_u^H$, $S' = v_d^S$, $B' = v_f^B$, we can obtain that

$$P_{r|ih\bullet} = P(\mathcal{H} = q_r^H | H' = v_i, X = \nu_h) = \frac{b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r})}{\sum_{r=1}^n (b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r}))} \quad (8)$$

Proof. By the theory of Bayesian networks, the joint probability distribution for the Bayesian network in Fig. 2 can be calculated using the formula:

$$\begin{aligned} & P(H, S, B, L, N, H', S', B', L', N') \\ &= P(B)P(S)P(B'|B)P(S'|S)P(H|B, S)P(L|H) \\ & \quad P(L'|L)P(N|H)P(N'|N)P(H'|H) \end{aligned}$$

Then we can calculate the probability $P_{r|ih\bullet}$ by the variable eliminating. Let

$$\begin{aligned} \varphi_1(L', H) &= \sum_L P(L|H)P(L'|L) \\ \varphi_2(N', H) &= \sum_N P(N|H)P(N'|N) \\ \varphi_3(B', S, H) &= \sum_B P(B)P(B'|B)P(H|B, S) \\ \varphi_4(S', B', H) &= \sum_S P(S)P(S'|S)\varphi_3(B', S, H) \\ \varphi_5(L', N', S', B', H', H) &= P(H'|H)\varphi_1(L', H) \\ & \quad \varphi_2(N', H)\varphi_4(S', B', H) \end{aligned}$$

Then the probability $P(\mathcal{H}, H', L', N', S', B')$ can be calculate as

$$\begin{aligned} P_{r|ih\bullet} &= P(\mathcal{H} = q_r^H, H' = v_i^H, X = \nu_h) \\ &= \varphi_5(\mathcal{H} = q_r^H, H' = v_i^H, X = \nu_h) \\ &= b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r}) \end{aligned} \quad (9)$$

The probability $P(H', L', N', S', B')$ can be calculated by summing up $P(\mathcal{H}, H', L', N', S', B')$.

$$\begin{aligned} P_{Tij\bullet} &= P(H' = v_i^H, X = \nu_h) = \sum_{r=1}^n P_{r|ih\bullet} = \\ & \sum_{r=1}^n (b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r})) \end{aligned} \quad (10)$$

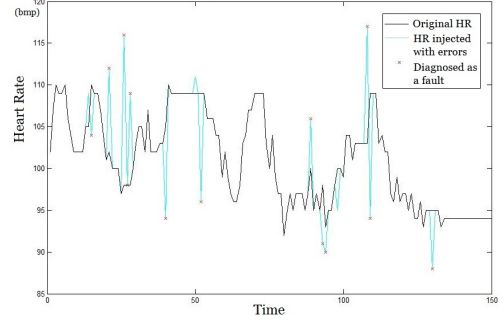


Fig. 3. The Simulation Result of the Fault Detection Method.

Then probability $P(\mathcal{H}|H', L', N', S', B')$ can be obtained

$$\begin{aligned} P_{r|ij\bullet} &= P(\mathcal{H} = q_r^H | H' = v_i^H, X = \nu_h) = \frac{P_{r|ij\bullet}}{P_{Tij\bullet}} \\ &= \frac{b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r})}{\sum_{r=1}^n (b_{ri}^H \sum_{l=1}^n (\bar{a}_{rl} b_{lj}^H) \sum_{l=1}^n (a_{rl} b_{lu}^H) \sum_{l=1}^k (\pi_l^S b_{ld}^S) \sum_{l'=1}^m (\pi_{l'}^P b_{lf}^P c_{l'l_r}))} \end{aligned} \quad (11)$$

Given an integer $1 \leq \theta \leq n'$, we define a function $\tilde{\theta} = \vartheta$ satisfying $1 \leq \vartheta \leq n$ and $q_{\tilde{\theta}}^H = v_{\theta}^H$. Given an integer $1 \leq \vartheta \leq n$, we define $\tilde{\vartheta} = \theta$ with $1 \leq \theta \leq n'$ and $q_{\tilde{\vartheta}}^H = v_{\theta}^H$. The probability of the estimated H equalling to the sensor reading H' for some $H' = v_{\theta}^H$ can be calculated as

$$P_{\theta|\theta j\bullet} = \frac{P_{\tilde{\theta}|\tilde{\vartheta} j\bullet}}{P_{T\theta j\bullet}} \quad (12)$$

By (12), we can give a statistic to decide whether the sensor reading H' is faulty by the evidence of its neighbour's reading S' , B' and its history and next time readings L' and N' , which uses a threshold $0 < \delta < 1$ as: if

$$P(\mathcal{H} = q_{\tilde{\theta}}^H | H' = v_{\theta}^H, V = \nu_j) > \delta \quad (13)$$

then $\mathcal{H}$ is set to $q_{\tilde{\theta}}^H$, and the sensor reading is diagnosed as correct. Otherwise, the sensor reading is considered as faulty.

Fig. 3 shows a simulation result using the proposed fault detection method. In this simulation, we select 10000 data with three attributes: BPmean, HR, and SpO2 from a real medical dataset. We use the first 9850 data to estimate the Bayesian network using the maximum likelihood estimation method. Then we inject 5% errors to the 150 observational BPmean and SpO2 data, and 10% errors to the 150 HR data. The original HR data, the HR data with injected errors, and the data diagnosed as faults are all marked in Fig. 3.

IV. ANALYSIS OF FAULT DETECTION

For a fault detection method, we can use three indicators to test its performance—the fault detection rate, the false alarm rate and the error rate after executing the fault diagnosis algorithm. In this section, we give theoretical analysis of these testing indicators of our proposed diagnosis method.

A. Performance Analysis of Fault Detection Method

Give an integer set $\{q_i^H | 1 \leq i \leq n\}$ and $1 \leq \theta \leq n$, we use $\lambda = q_{\theta}^H$ to denote $\bigvee_{1 \leq \theta \neq \theta \leq n} \lambda = q_{\theta}^H$. In order to calculate the theoretical average fault detection and false alarm rates, the following probabilities are important.

$$P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H)$$

$$P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H)$$

To calculate these two probabilities, let's first consider the probability $P(L', N', S', B' | H', H)$. Let

$$P_{ri} = P(H = q_r^H, H' = v_i^H) = \sum_{h=1}^{(n')^2 m' k'} P_{rih\bullet} \quad (14)$$

Then we can obtain that

$$P(X = \nu_h | H' = v_i^H, H = q_r^H) = \frac{P(H = q_r^H, H' = v_i^H, X = \nu_h)}{P(H' = v_i^H, H = q_r^H)} = \frac{P_{rih\bullet}}{P_{ri}} \quad (15)$$

The probability $P(\mathcal{H} | H', H)$ can be calculated by $P(\mathcal{H} | H', L', N', S', B')$ and $P(L', N', S', B' | H', H)$ as follows.

$$\begin{aligned} P_{\kappa|ir} &= P(\mathcal{H} = q_{\kappa}^H | H' = v_i^H, H = q_r^H) \\ &= \sum_{h=1}^{(n')^2 m' k'} P(\mathcal{H} = q_{\kappa}^H | H' = v_i^H, H = q_r^H, X = \nu_h) \\ &\quad P(X = \nu_h | H' = v_i^H, H = q_r^H) \\ &= \sum_{h=1}^{(n')^2 m' k'} P_{\kappa ih\bullet} \frac{P_{rih\bullet}}{P_{ri}} \end{aligned} \quad (16)$$

As a special case, $P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H)$ and $P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H)$ can be derived as follows.

$$\begin{aligned} P_{\theta|\theta\theta} &= P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &= \sum_{h=1}^{(n')^2 m' k'} P_{\theta|\theta h\bullet} \frac{P_{\theta\theta h\bullet}}{P_{\theta\theta}} \end{aligned} \quad (17)$$

$$\begin{aligned} P_{\theta|\hat{\theta}\theta} &= P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &= \sum_{h=1}^{(n')^2 m' k'} P_{\theta|\hat{\theta}h\bullet} \frac{P_{\theta\hat{\theta}h\bullet}}{P_{\theta\hat{\theta}}} \end{aligned} \quad (18)$$

Now we can calculate the theoretical average rate σ of sensor faults corrected by the proposed approach.

$$\begin{aligned} \sigma &= \sum_{\theta=1}^n \sum_{\theta=1, \theta \neq \hat{\theta}}^{n'} P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &\quad P(H' = v_{\theta}^H | H = q_{\theta}^H) P(H = q_{\theta}^H) \\ &= \sum_{\theta=1}^n \sum_{\theta=1, \theta \neq \hat{\theta}}^{n'} b_{\theta\theta}^H \pi_{\theta}^H P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &= \sum_{\theta=1}^n \sum_{\theta=1, \theta \neq \hat{\theta}}^{n'} b_{\theta\theta}^H \pi_{\theta}^H (1 - P_{\theta|\theta\theta}) \end{aligned} \quad (19)$$

Although the Bayesian network based fault diagnosis method can detect sensor faults, it may also introduce new errors. This

effect can be measured by the theoretical false alarm rate ϕ .

$$\begin{aligned} \phi &= \sum_{\theta=1}^n P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &\quad P(H' = v_{\theta}^H | H = q_{\theta}^H) P(H = q_{\theta}^H) \\ &= \sum_{\theta=1}^n b_{\theta\theta}^H \pi_{\theta}^H P(\mathcal{H} = q_{\theta}^H | H' = v_{\theta}^H, H = q_{\theta}^H) \\ &= \sum_{\theta=1}^n b_{\theta\theta}^H \pi_{\theta}^H (1 - P_{\theta|\hat{\theta}\theta}) \end{aligned} \quad (20)$$

And the expected rate of decoding errors μ after executing this fault detection algorithm can be calculated as follows.

$$\begin{aligned} \mu &= p(1 - \sigma) + (1 - p)\phi \\ &= p(1 - \sum_{\theta=1}^n \sum_{\theta=1, \theta \neq \hat{\theta}}^{n'} b_{\theta\theta}^H \pi_{\theta}^H (1 - P_{\theta|\theta\theta})) + \\ &\quad (1 - p) \sum_{\theta=1}^n b_{\theta\theta}^H \pi_{\theta}^H (1 - P_{\theta|\hat{\theta}\theta}) \end{aligned} \quad (21)$$

B. The Decision Threshold Scheme

The decision threshold scheme is to select the optimal threshold minimizing μ . Given a CPT of the Bayesian network and a threshold δ , we define $\varrho_{r|ij}$ as

$$\varrho_{r|ij\bullet} = \begin{cases} 1, & \text{if } P_{r|ij\bullet} > \delta \\ 0, & \text{otherwise} \end{cases}$$

We use $\mu(P/\varrho)$ to denote the equation obtained from (21) by replacing $P_{\theta|\theta h\bullet}$ and $P_{\theta|\hat{\theta}h\bullet}$ respectively by $\varrho_{\theta|\theta h\bullet}$ and $\varrho_{\theta|\hat{\theta}h\bullet}$. Then the decision threshold scheme is to find a threshold δ that minimizes $\mu(P/\varrho)$, written as:

$$\delta_{op} = \min_{\delta} \mu(P/\varrho) \quad (22)$$

We give the following Algorithm 1 for the approximate discrete decision threshold scheme, which is to find the optimal threshold δ from a discrete threshold set Θ that minimizes $\mu(P/\varrho)$.

Algorithm 1 Discrete Decision Threshold Scheme:

Input: A probability distribution set of a Bayesian network and a discrete threshold set Θ

Output: The optimal threshold δ_{op}

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 $\delta_{op} = 0;$ 
 $\min = 1;$ 
for each  $\delta \in \Theta$  do
  if  $\mu(P/\varrho) < \min$  then
     $\min = \mu(P/\varrho);$ 
     $\delta_{op} = \delta;$ 
  end if
end for
return  $\delta_{op};$ 

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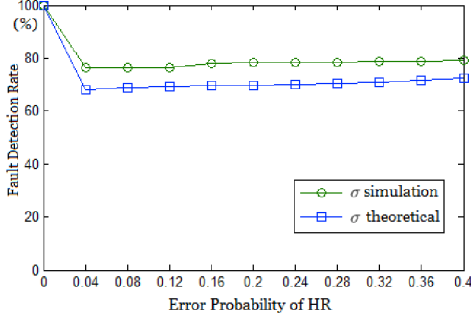


Fig. 4. The Average Fault Detection Rate with the Approximate Optimal Threshold.

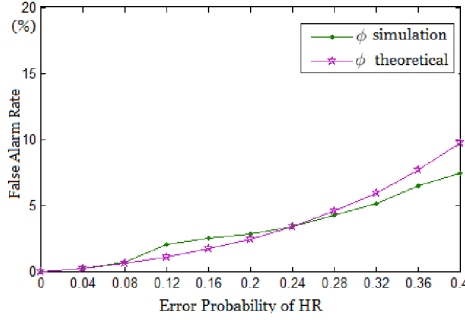


Fig. 5. The Average False Alarm Rate with the Approximate Optimal Threshold.

V. EXPERIMENT RESULTS

In order to examine the performance of the proposed fault diagnosis method, we carry out some experiments on medical datasets from the PhysioNet database (MIMIC Database) [15]. The dataset contains 7 attributes: BPmean, systolic BP, diastolic BP, HR, pulse, respiration rate, and SpO2. We only focus on three attributes: BPmean, HR, and SpO2. In our experiment, we use a data sequence of 10000 data. We select the first 9000 data as the training data, and the last 1000 data with injecting faults as diagnosed data.

Since the 9000 training data are not injected faults, so given sensor error probabilities p_h , p_b and p_s for HR, BPmean and SpO2, we can construct a Bayesian network by the maximum likelihood estimation approach. Using Algorithm 1, we can calculate the theoretical average fault detection rate σ , false alarm rate ϕ and the expected error rate μ after executing this fault diagnosis algorithm with the approximate optimal threshold. Finally, given sensor error probabilities, we inject faults into the 1000 diagnosed data with the position and the value of the injected error all selected by random numbers. The simulation results are the average performance of 300 times randomized experiments.

We fixed the error probabilities p_b and p_s of BPmean and SpO2 at 5%, Fig. 4-6 show the performance of the proposed method with the approximate optimal threshold decision scheme of both the simulation and the theoretical prediction.

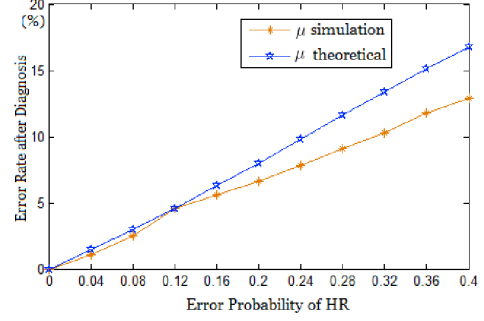


Fig. 6. The Average Error Rate after Executing the Diagnosis Algorithm with the Approximate Optimal Threshold.

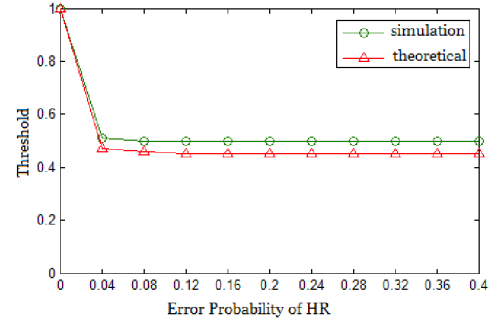


Fig. 7. The Approximate Optimal Decision Threshold.

The simulation performance of the diagnosis process depends on the sensor error probability and the threshold δ . Besides, the range in which the injected errors must lie is another factor influencing the diagnosis performance, because the closer the fault to the ground true value, the harder it is to be detected. In our simulation experiments, we select the same data range of injected errors to the normal value of physiological signs. The conclusion from these results is that the simulation matches the theoretical predictions in all respects. Moreover, these figures show that the number of errors can be reduced by over 60% after executing our fault diagnosis method.

As we know, the difficulty of the threshold based fault diagnosis method is how to determine a threshold. Through our analysis, we can determine the approximate optimal threshold by Algorithm 1, Fig. 7 shows the approximate optimal thresholds obtained respectively by Algorithm 1 and simulation. We can see that the simulation and theoretical prediction optimal threshold match closely.

The simulation result in Fig.4 tells us that around 80% of the total faults can be corrected, which is not very high. The reason is that the probability σ of sensor faults corrected is influenced by the threshold δ , the larger of δ , the higher of σ and ϕ . Fig. 8 shows the simulation result by fixed $\delta = 0.8$, in which the probability σ of sensor faults detected increases obviously than which in the simulation with the approximate optimal threshold scheme.

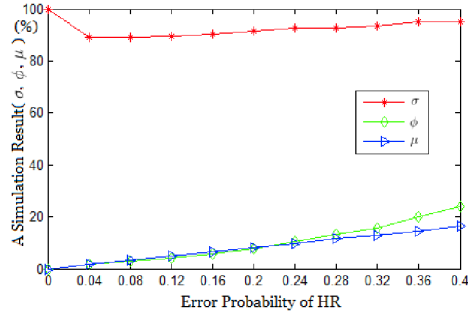


Fig. 8. The Simulation Performance with the Threshold Fixed as 0.8.

VI. CONCLUSIONS

BSNs are important techniques used for the health person monitoring. However, due to the restriction of resources, medical sensors usually produce unreliability measurements. So fault detection is very important for BSNs to avoid false medical diagnosis and false alarms. In this paper, we use Bayesian network based method for fault detection of BSNs. We formalize a Bayesian network model to describe the body sensor network, and give a method to decide whether a sensor reading is faulty or not. Then we do theoretical analysis of the fault detection, false alarm and the error probability after executing the fault diagnosis algorithm, and give an approximate algorithm for the discrete decision threshold scheme. Finally, our experiments show that the simulation matches the theoretical predictions, and the number of errors can be reduced by 60% after executing our fault diagnosis method.

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